10.Applying Performance Optimization Techniques (Indexing, Subforms, and Query Tuning) in Zoho Creator

Aim:

To implement and apply performance optimization techniques - specifically Field Indexing, Subform structuring, and Deluge Query Tuning - in the Patient Appointment & Scheduling Management application developed using Zoho Creator, to improve appointment record retrieval, reduce response latency, and optimize database operations.

Overview:

As the Patient Appointment & Scheduling Management application handles more bookings, appointments, doctors, and follow-up records, inefficient form structures or broad Deluge queries may increase response time. Three relevant optimization approaches are:

1. **Field Indexing:**
   Concept: Identify fields used frequently in searches, such as Email Address, Preferred Appointment Date, Doctor Name, and Status. Database indexing may improve lookups where Creator exposes or manages such indexes.
   Benefit: Faster targeted searches and filters where suitable indexes are supported. Marking an email field as unique is not appropriate because one patient can book multiple appointments.
2. **Efficient Subform Usage:**
   Concept: Store several requested appointment services as child rows within the parent Appointment_Booking_Form by using an Appointment Services subform.
   Benefit: Related service details can be entered together without duplicating the whole booking record.
3. **Deluge Query Tuning & Best Practices:**
   Concept: Apply selective criteria, sorting, and range limits to appointment queries instead of retrieving every record or performing repeated reads within loops.
   Benefit: Reduces unnecessary record retrieval and helps improve workflow efficiency.

Procedure

Step 1: Identifying and Applying Field Indexing

1. Open Patient Appointment & Scheduling Management in the Zoho Creator application builder.
2. Navigate to Design > Appointment_Booking_Form and select a frequently queried field, such as Email Address or Preferred Appointment Date.
3. Review the field Properties and available Advanced / Validation settings for a supported indexing option.
4. Where no indexing control is available, document the selected search fields; do not enable uniqueness on patient email, because repeat appointments use the same address.

Step 2: Structuring and Optimizing Subforms

1. Open Appointment_Booking_Form in the form builder and add a Subform named Appointment Services (link name: Appointment_Services).
2. Add essential child fields such as Service Name; optionally add Service Notes and Estimated Duration if needed for the booking process.
3. Keep the subform optional and limit its maximum entries to 5 to reduce unnecessary rows and form payload size.

Step 3: Writing Optimized Deluge Scripts (Query Tuning)

1. Open Workflows > Functions and create the function Optimized_Appointment_Query in the Deluge editor.
2. Avoid broad queries that retrieve every appointment record. Use the Preferred_Appointment_Date condition and a bounded result range.

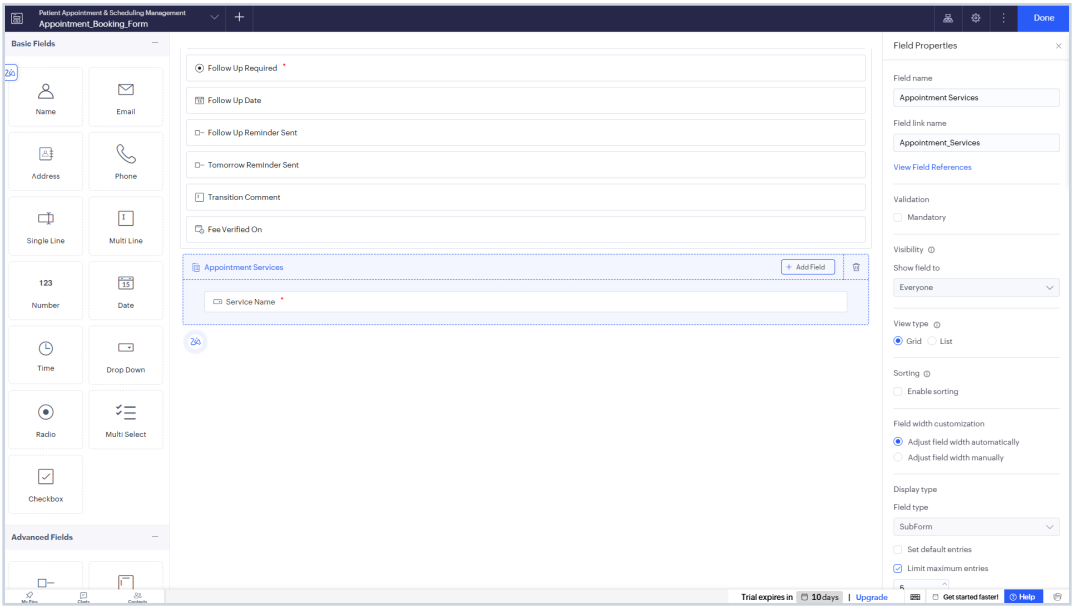
```
appointments = Appointment_Booking_Form
[Preferred_Appointment_Date >= zoho.currentdate]
sort by Preferred_Appointment_Date
range from 1 to 50;
info "Fetched appointments: " + appointments.count();
```

3. Save and execute the function where supported. Avoid repeated record fetches inside loops.

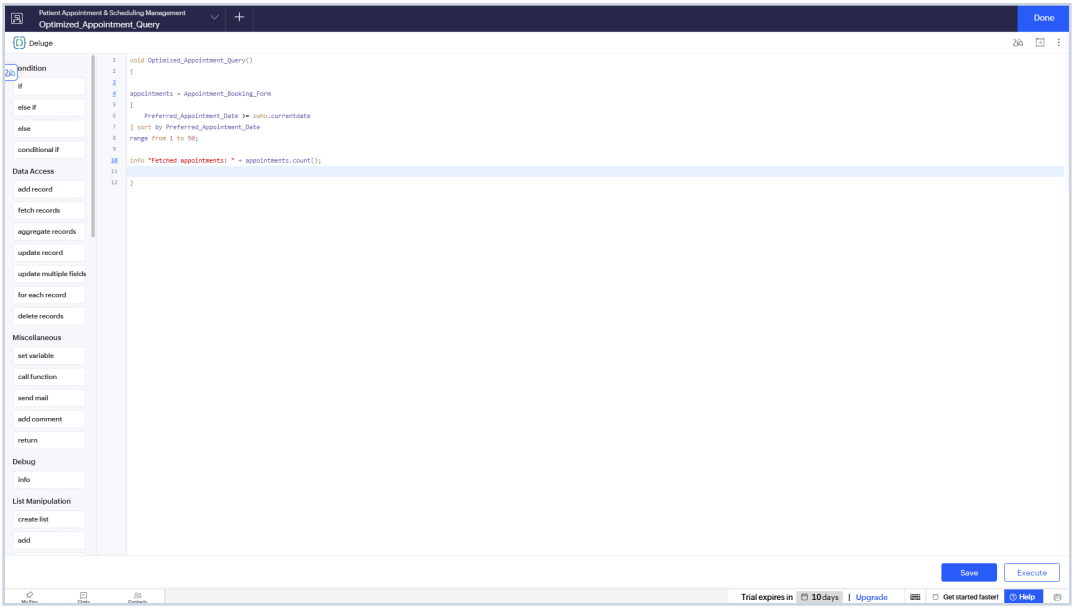
Step 4: Testing and Monitoring Performance

1. Use Execute, the Deluge output, or available execution logs to inspect the fetched appointment count.
2. Test appointment report loading and form submission with sample records, comparing equivalent before-and-after conditions when measurements are available.

Screenshots



Appointment Services subform in Appointment_Booking_Form



Optimized_Appointment_Query in the Deluge editor

Result

The Appointment Services subform was structured with an entry limit, and a targeted Deluge appointment query was prepared using date criteria, sorting, and a 50-record range. These changes demonstrate subform organization and query tuning in the Patient Appointment & Scheduling Management application. Field indexing and measured speed improvements remain subject to feature availability and performance testing.